Lesson notes · for you

Data quality and contracts

Data engineering · Level 4 · Lesson 3

Time
17 min

Materials
Your phone or a computer, for the SQL and Python steps; Paper, for writing a contract

Learning goals

- Write quality tests as code: types, nulls, ranges, uniqueness, references, freshness.
- Agree a data contract between producers and users.
- Decide what a failed test does.

Activity: Contract and tests

1. Choose (3 min): Pick a dataset you depend on, produced by someone else.
2. Contract (6 min): Write the columns, types, allowed values, the key and the freshness rule.
3. Test (5 min): Turn three rules into tests that return the bad rows.
4. Decide (3 min): For each test, choose what a failure does, and why.

Swap contracts with a partner and try to write a row that breaks theirs but passes every test.

Discussion

- Why use a LEFT JOIN to find orphans?
- When is quarantine better than blocking the load?
- What can't any test catch?

Check yourself

- Write one rule of a data contract and the SQL test that checks it.

Try it this week
Write a data contract for a dataset you use, like your class timetable or a club's results. List the columns, types and allowed values, and who fixes it when it's wrong.

Worksheet

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Name _____ Class _____ Date _____

1 **A test fails on 3 rows out of 10,000. Which response keeps the pipeline moving and the data honest?**

- ◯ Quarantine the 3 rows and alert the producer
- ◯ Block the whole load until next week
- ◯ Let them through and say nothing

2 **Match each side of quality to a fault that breaks it.**

Valid	• • • •	Monday's file still missing on Wednesday
Unique	• • • •	A mark for a pupil who doesn't exist
Consistent	• • • •	The same pupil's mark loaded twice
Timely	• • • •	A mark of 25 out of 20

3 **A referential integrity test: find every mark whose pupil_id has no row in pupils. Show the pupil_id, subject and mark of each orphan.**

```
SELECT marks.pupil_id, subject, mark FROM marks
JOIN pupils ON pupils.id = marks.pupil_id
WHERE pupils.id IS NULL;
```

Write your program:

4 **The contract says: subject is maths or science, and mark is a whole number from 0 to 20. problems(row) returns a list of reasons a row breaks it, but it misses one rule. Fix it, so the program prints every failure with its line number.**

```
import csv
SUBJECTS = ["maths", "science"]
def problems(row):
    out = []
    if row["subject"] not in SUBJECTS:
        out.append("unknown subject")
    if not row["mark"].isdigit():
        out.append("mark not a number")
    return out
with open("marks.csv") as f:
    for n, row in enumerate(csv.DictReader(f), 2):
        for p in problems(row):
            print("line", n, p)
```

Write your program:

5 **This range test flags 9 as out of range, which it isn't. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  for row in [{"mark": "9"}, {"mark": "25"}, {"mark": "14"}]:
2      if row["mark"] > "20":
3          print(row["mark"], "out of range")
```

✎ **Which dataset do you depend on that someone else produces (a timetable, results, a club list)? What would you put in its contract?**